Lab 13: AC Circuits - Phase Relationships

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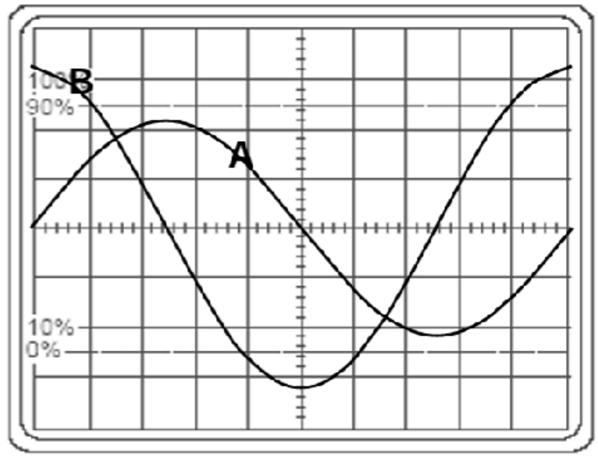
Pre-lab Questions

1. *In the figure to the right, does curve A lead or lag curve B?*

   Curve B leads curve A since B reaches its peak before curve A does.



2. *The oscilloscope vertical scale is set to* 5 V/div *and the horizontal scale is set to* 5 ms/div.

   a) *What is the time difference in the curves?*

   Curve A reaches its peak approximately 2.5 divisions after curve B does, so:

   $$\Delta t \approx (2.5\text{ div})(5\text{ ms/div}) = \mathbf{12.5\,ms}.$$

   b) *What is the phase difference in the curves?*

   Assuming both curves have the same frequency, curve B leads A by approximately:

   $$\Delta\varphi = \mathbf{90^\circ = \pi/2\ radians}.$$

   This is because curve B reaches its peak approximately a quarter of a period before curve A does. More precisely, if $t = 0$ was fixed on the left side of the screen, curve B roughly models $B_0 \cos(\omega t)$ whereas A models $A_0 \sin(\omega t) = A_0 \cos(\omega t - \pi/2)$.

3. A resistor, inductor, and capacitor are connected in series and the potential difference across each are measured as $V_R = 35$ V, $V_L = 45$ V, and $V_C = 55$ V respectively. What is the input voltage V?

   The impedance of the circuit is given by:

   $$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}.$$

   By "Ohm's law for impedance", the potential difference across the LRC components is given by:

   $$\begin{aligned}\sum V = IZ &= I\sqrt{R^2 + \left(X_L - X_C\right)^2} \\ &= \sqrt{I^2R^2 + I^2\left(X_L - X_C\right)^2} \\ &= \sqrt{(IR)^2 + \left(IX_L - IX_C\right)^2} \\ &= \sqrt{V_R^2 + \left(V_L - V_C\right)^2}.\end{aligned}$$

   By Kirchhoff's voltage law, the input voltage V must be equal to the sum of the potential differences across the components, so:

   $$V = \sqrt{V_R^2 + \left(V_L - V_C\right)^2} = \sqrt{(35\text{ V})^2 + (45\text{ V} - 55\text{ V})^2} = \mathbf{36.4\,V}.$$

Abstract

In this experiment, an AC source drove series RC and LR circuits at various frequencies ω, and the phase relationships between voltages across the components were observed and verified using an oscilloscope and digital multimeter (DMM).

After finding a suitable frequency f_0, the RC circuit was driven at frequencies between $0.25f_0$ and $5f_0$. Phase angles calculated by observing time differences displayed on the oscilloscope agreed with values computed from the formula $\cos\theta = V_R/V_S$ (where voltages were measured using a DMM) within a few degrees at each frequency. A linear regression of $\tan\theta$ versus $1/\omega$ yielded an extracted capacitance and resistance within 25% of the labeled values. The Pythagorean relation $V_S \approx \sqrt{V_R^2 + V_C^2}$ was verified at each frequency. The LR circuit was similarly analyzed, confirming $\tan\theta = \omega L/R$ and yielding an extracted inductance of ≈ 21.5 mH (similar to the labeled value of 22 mH).

Introduction

The purpose of this lab is to observe and quantify phase relationships between voltages in AC series circuits containing a resistor (R), capacitor (C), and inductor (L). When an AC source $V_S = V_0\cos(\omega t)$ drives a series RC circuit, the current leads the source voltage by a phase angle θ such that:

$$\tan\theta = \frac{X_C}{R} = \frac{1}{\omega RC},$$

where $X_C = 1/(\omega C)$ is the capacitive reactance. Because the resistor voltage V_R is in phase with the current, this phase angle also satisfies $\cos\theta = V_R/V_S$ and $\tan\theta = V_C/V_R$. The relation $V_S = \sqrt{V_R^2 + V_C^2}$ follows from the phasor diagram. Similarly, for a series LR circuit the source voltage leads the current by:

$$\tan\theta_L = \frac{X_L}{R} = \frac{\omega L}{R},$$

so $\tan\theta_L = V_L/V_R$. In both cases the phase can be measured either directly from the oscilloscope time-delay Δt or indirectly from voltage ratios measured with a DMM.

Equipment

- Oscilloscope
- Function Generator
- Digital Multimeter (DMM)
- Resistors (1000 Ω, 1500 Ω), Capacitor (0.10 μF), and Inductor (22 mH)
- Miscellaneous: Wires, Oscilloscope Alligator Clip Probes, BNC cables, etc.

Part 1: RC Series Circuit

Labeled Resistance of Resistor	$R = 1000\,\Omega$
Labeled Capacitance of Capacitor	$C = 0.10\,\mu F$
Frequency at which $V_R \approx V_S/2$	$f_0 = 976\,\text{Hz}$

We selected frequencies between $0.25 f_0 = 244$ Hz and $5 f_0 = 4880$ Hz. At each frequency, the time difference Δt was observed from the oscilloscope, and the phase angle[1] θ_0 was calculated using:

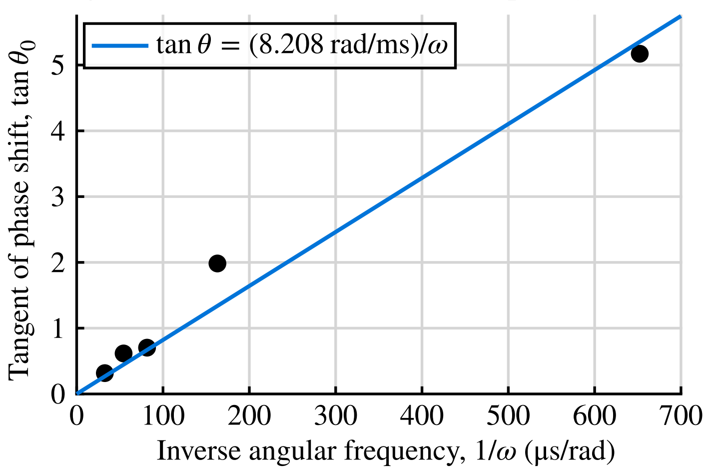
$$\theta_0 = \omega \Delta t = 2\pi f \Delta t.$$

Phase Measurements using Oscilloscope in RC series circuit

Frequency f	**$1/\omega = 1/(2\pi f)$**	**Time shift Δt**	**Phase $\theta_0 = 2\pi f \Delta t$**	**$\tan\theta_0$**
244.0 Hz	652 µs/rad	0.9 div · 1000 µs/div = 900 µs	79.1°	5.17
976.0 Hz	163 µs/rad	0.9 div · 200 µs/div = 180 µs	63.2°	1.98
1952 Hz	81.5 µs/rad	0.5 div · 100 µs/div = 50 µs	35.1°	0.704
2928 Hz	54.4 µs/rad	0.6 div · 50 µs/div = 30 µs	31.6°	0.616
4880 Hz	32.6 µs/rad	0.2 div · 50 µs/div = 10 µs	17.6°	0.317

At $f = f_0$, the phase difference between the resistor and voltage source was approximately $\theta_0 = 63.2°$, which is close to the expected value of $\theta = 60°$.

Tangent of Phase Shift as Inverse Frequency Varies



The slope of the regression line is 8.208 rad/ms. It can be seen that:

$$\tan\theta_0 = \frac{X_C}{R} = \frac{1}{\omega RC} = \frac{1}{RC}\frac{1}{\omega}.$$

Thus, the slope of the regression line is equal to $1/(RC)$, so we can extract the capacitance C as:

[1] Between the resistor and the AC source.

$$\text{slope} = \frac{1}{RC} \quad \rightarrow \quad C = \frac{1}{R \cdot \text{slope}} = \frac{1}{(1000\,\Omega)(8.208\,\text{rad/ms})} = 0.098\,\mu\text{F}.$$

This is in agreement with the labeled capacitance of $C = 0.10\,\mu\text{F}$ (2.43 % error.)

Similarly, we can extract the resistance R using a similar method:

$$R = \frac{1}{C \cdot \text{slope}} = \frac{1}{(0.1\,\mu\text{F})(8.208\,\text{rad/ms})} = 1218\,\Omega.$$

This is in agreement with the labeled resistance of $R = 1000\,\Omega$ (21.8 % error.)

DMM Measurements

For each of the frequencies, the RMS of the voltage across the capacitor V_C and resistor V_R were measured using a DMM. The RMS voltage of the generator V_S was also measured.

Using the relationship:

$$\cos\theta_1 = \frac{V_R}{V_S} \quad \rightarrow \quad \theta_1 = \cos^{-1}\left(\frac{V_R}{V_S}\right),$$

the phase difference θ_1 between the resistor and voltage source was calculated at each frequency. This θ_1 was compared to the measured θ_0 from the oscilloscope.

Phase Measurements from DMM voltage ratios in RC series circuit

Frequency f	V_S	V_C	V_R	$\theta_1 = \cos^{-1}(V_R/V_S)$	θ_0	**% Difference**
244.0 Hz	2.78 V	2.74 V	0.410 V	81.5°	79.1°	3.1%
976.0 Hz	2.72 V	2.30 V	1.37 V	59.8°	63.2°	5.7%
1952 Hz	2.66 V	1.69 V	2.00 V	41.2°	35.1°	15.0%
2928 Hz	2.61 V	1.26 V	2.24 V	30.9°	31.6°	2.4%
4880 Hz	2.55 V	0.740 V	2.39 V	20.4°	17.6°	14.9%

V_S, V_C, and V_R are the RMS voltages measured using a DMM. θ_1 is the phase angle calculated from the voltage ratios, and θ_0 is the phase angle calculated from the oscilloscope time shift from the previous table. Δt was excluded, since θ_0 was calculated from Δt.

The direct observation of phase shift from the oscilloscope and the indirect calculation of phase shift from voltage ratios measured with a DMM agreed within $\leq 15\%$ for all frequencies.

The relation $V_S \approx \sqrt{V_R^2 + V_C^2}$, can be verified as follows (all % errors are $< 2\%$):

Frequency f	V_C	V_R	**Measured V_S**	$\sqrt{V_R^2 + V_C^2}$	**% Error**
244.0 Hz	2.74 V	0.410 V	2.78 V	2.77 V	0.3%
976.0 Hz	2.30 V	1.37 V	2.72 V	2.68 V	1.6%
1952 Hz	1.69 V	2.00 V	2.66 V	2.62 V	1.6%
2928 Hz	1.26 V	2.24 V	2.61 V	2.57 V	1.5%
4880 Hz	0.740 V	2.39 V	2.55 V	2.50 V	1.9%

Part 2: LR Series Circuit

Labeled Inductance of Inductor	$L = 22$ mH
Labeled Resistance of Resistor	$R = 1500\,\Omega$

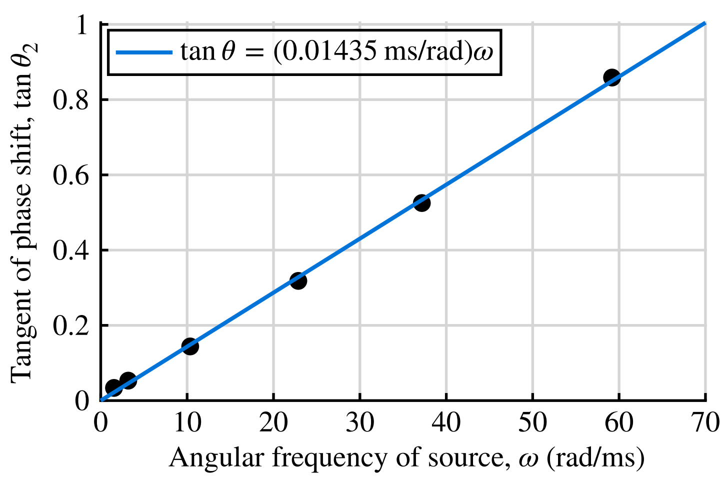
Using a DMM, the RMS voltage across the inductor V_L and resistor V_R were measured at various frequencies. The phase difference between the resistor and voltage source θ_2 was calculated using:

$$\tan\theta_2 = \frac{V_L}{V_R} \quad \rightarrow \quad \theta_2 = \tan^{-1}\left(\frac{V_L}{V_R}\right).$$

Phase Measurements from DMM voltage ratios in LR series circuit

Frequency f	$\boldsymbol{\omega = 2\pi f}$	$\boldsymbol{V_L}$	$\boldsymbol{V_R}$	$\boldsymbol{\tan\theta_2 = V_L/V_R}$	$\boldsymbol{\theta_2 = \tan^{-1}(V_L/V_R)}$
244.0 Hz	1.533 rad/ms	0.0887 V	2.63 V	0.0337	1.93°
507.0 Hz	3.186 rad/ms	0.139 V	2.62 V	0.0531	3.04°
1648 Hz	10.35 rad/ms	0.370 V	2.57 V	0.144	8.19°
3640 Hz	22.87 rad/ms	0.770 V	2.42 V	0.318	17.7°
5916 Hz	37.17 rad/ms	1.15 V	2.19 V	0.525	27.7°
9420 Hz	59.19 rad/ms	1.58 V	1.84 V	0.859	40.7°

Tangent of Phase Shift as Frequency Varies in LR Circuit



The slope of the regression line is 0.01435 ms/rad. Since:

$$\tan\theta_2 = \frac{X_L}{R} = \frac{\omega L}{R},$$

the slope of the regression line is equal to L/R, so we can extract the inductance L as:

$$L = R \cdot \text{slope} = (1500\,\Omega)(0.01435\text{ ms/rad}) = 21.53\text{ mH}.$$

This is in agreement with the labeled inductance of $L = 22$ mH (2.15 % error.)

Similarly, we can extract the resistance R using a similar method:

$$R = \frac{L}{\text{slope}} = \frac{22\,\text{mH}}{0.01435\,\text{ms/rad}} = 1533\,\Omega.$$

This is in agreement with the labeled resistance of $R = 1500\,\Omega$ (2.19 % error.)

Equal-Amplitude Frequency with Oscilloscope

At $f = 10\,470\,\text{Hz}$, the voltage across the inductor and resistor were approximately equal.

The time difference between the curves on the oscilloscope was approximately $\Delta t = 1.1\,\text{div} \cdot 20\,\mu\text{s/div} = 22\,\mu\text{s}$, which corresponds to a phase difference of $\theta = 2\pi f \Delta t = 82.9°$. This is close to the expected value of $\theta = 90°$ for equal amplitudes across the inductor and resistor (7.86% error).

The phase angle is not *exactly* 90° due to the internal resistance of the wires (e.g. the inductor is made up of spun-up wire which has some resistance), which causes the inductor voltage to be slightly out of phase with the current.

Conclusion

Error Analysis

Sources of uncertainty which may have affected the results include:

- **Internal resistance of inductor**, since we assumed the inductor as an ideal inductor with no internal resistance. This may have caused the phase difference between voltage and current across the inductor to be less than $\pi/2$, which would affect the accuracy of our measurements and calculations for the LR circuit.
- **Human error when measuring Δt**, since the time difference between the curves on the oscilloscope was "eyeballed" which would introduce random error in Δt and thus in the calculated phase angle θ.
- **Uncertainty in component measurements**, since we took the labeled values of R, C, and L in the resistor, capacitor, and inductor at face value without measuring them directly, which may have introduced systematic error if the actual values of the components differed from their labeled values.

Questions

1. Starting with definitions for resistance R and capacitance C, show that RC has units of time.

   The units of R and C are given by:

$$\Delta V = IR \quad \rightarrow \quad R = \frac{\Delta V}{I} \quad \rightarrow \quad [R] = \frac{[\Delta V]}{[I]} = \frac{\mathrm{V}}{\mathrm{A}} = \frac{\mathrm{V}}{\mathrm{C/s}} = \frac{\mathrm{V\,s}}{\mathrm{C}}.$$
$$C = \frac{Q}{\Delta V} \quad \rightarrow \quad [C] = \frac{[Q]}{[\Delta V]} = \frac{\mathrm{C}}{\mathrm{V}}.$$

   Thus, the units of RC are given by:

$$[RC] = [R][C] = \frac{\mathrm{V\,s}}{\mathrm{C}}\frac{\mathrm{C}}{\mathrm{V}} = \mathrm{s} = [t].$$

2. Starting with definitions for R, C, and inductance L, show that $X_L = \omega L$ and $X_C = 1/(\omega C)$ have units of resistance, where ω is angular frequency ($[\omega] = \mathrm{rad/s} \equiv 1/\mathrm{s}$).

   The units of L are given by:

$$\mathcal{E} = L\frac{\mathrm{d}I}{\mathrm{d}t} \quad \rightarrow \quad L = \frac{\mathcal{E}}{\mathrm{d}I/\mathrm{d}t} \quad \rightarrow \quad [L] = \frac{[\mathcal{E}]}{[\mathrm{d}I/\mathrm{d}t]} = \frac{\mathrm{V}}{\mathrm{A/s}} = \frac{\mathrm{V}}{\mathrm{C/s^2}} = \frac{\mathrm{V\,s^2}}{\mathrm{C}}.$$

   The units of inductive reactance X_L are given by:

$$\left[X_L\right] = [\omega L] = [\omega][L] = \frac{\mathrm{rad}}{\mathrm{s}}\frac{\mathrm{V\,s^2}}{\mathrm{C}} \equiv \frac{1}{\mathrm{s}}\frac{\mathrm{V\,s^2}}{\mathrm{C}} = \frac{\mathrm{V\,s}}{\mathrm{C}} = \frac{\mathrm{V}}{\mathrm{C/s}} = \frac{[\Delta V]}{[I]} = \left[\frac{\Delta V}{I}\right] = [R].$$

   The units of capacitive reactance X_C are given by:

$$\left[X_C\right] = \left[\frac{1}{\omega C}\right] = \frac{1}{[\omega][C]} = \frac{1}{\frac{\mathrm{rad}}{\mathrm{s}}\frac{\mathrm{C}}{\mathrm{V}}} \equiv \frac{1}{\frac{1}{\mathrm{s}}\frac{\mathrm{C}}{\mathrm{V}}} = \frac{\mathrm{V\,s}}{\mathrm{C}} = \frac{\mathrm{V}}{\mathrm{C/s}} = \frac{[\Delta V]}{[I]} = \left[\frac{\Delta V}{I}\right] = [R].$$

3. Show that taking the time derivative of *any* sinusoidal function, such as $\cos(\omega t + \theta)$, has the effect of advancing its phase by $\pi/2$.

   Let $F(t) = \cos(\omega t + \theta)$ be a sinusoidal function. Evaluating the time derivative of f gives:

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\cos(\omega t + \theta) = -\omega \sin(\omega t + \theta).$$

   By trigonomic identities, we can show that $\cos(\alpha - \pi/2) = \sin(\alpha)$:

$$\cos\left(\alpha - \frac{\pi}{2}\right) = \cos(\alpha)\cos\left(\frac{\pi}{2}\right) + \sin(\alpha)\sin\left(\frac{\pi}{2}\right) = 0 + \sin(\alpha) = \sin(\alpha).$$

   Applying this identity to $\mathrm{d}F/\mathrm{d}t$, we have:

$$\frac{\mathrm{d}F}{\mathrm{d}t} = -\omega \sin(\omega t + \theta) = -\omega \cos\left((\omega t + \theta) - \frac{\pi}{2}\right).$$

   This shows that $\mathrm{d}F/\mathrm{d}t$ is a sinusoidal function with the same frequency as F but with its phase advanced (or delayed) by $\pi/2$. ∎

4. If the internal resistance of an inductor is not negligible, how does it change the relative phase of voltage and current for the inductor? Is this effect more important at high or at low frequency?

   If the internal resistance of an inductor is not negligible, then the inductor can be modeled as a resistor and an ideal inductor connected in series. The voltage across the inductor will have two components: one across the ideal inductor and one across the internal resistance. The voltage across the ideal inductor will lead the current by $\pi/2$, while the voltage across the internal resistance will be in phase with the current. Thus, the overall voltage across the inductor will lead the current by an angle less than $\pi/2$.

   This effect is more important at high frequency because the inductive reactance $X_L = \omega L$ increases with frequency, while the internal resistance remains constant. As a result, the voltage across the ideal inductor will "dominate" at high frequency, and the phase difference between voltage and current will be closer to $\pi/2$. At low frequency, the internal resistance will have a more significant effect on the overall voltage across the inductor, and the phase difference between voltage and current will be less than $\pi/2$. ∎